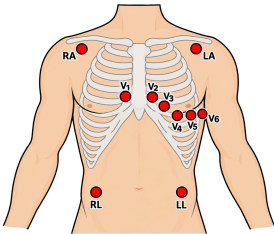


Diagnosis-Specific Selective Prediction for Safe Reduced Lead ECG AI

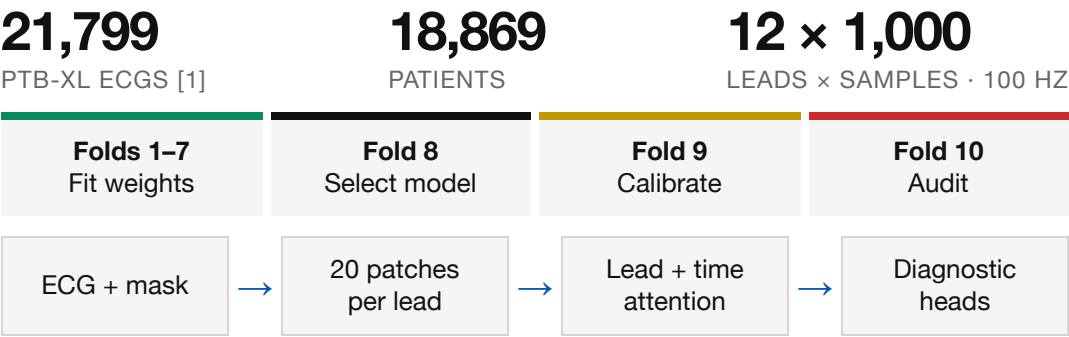


01 THE CLINICAL GAP

Portable ECG expands access, but average accuracy does not show which decisions are safe to automate.

02 DATA & MODEL COMPARISON

Control the input. Separate the patients.



Matched comparison: fixed I + II, random-lead and fixed 12-lead models use the same transformer, data and seed. Fixed and random test on I + II; rescue uses all 12 leads.

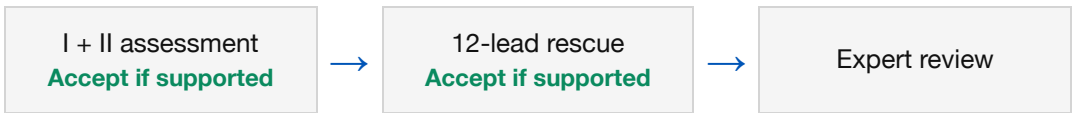
RESEARCH QUESTION

Can leads I + II rule out AFIB or AFLT with accepted-negative error below 2%?

03 FROM PROBABILITY TO AN ACTION

Require evidence before accepting a call.

Risk = P(disease present | accepted negative). AFIB/AFLT rule-out: validation upper error bound $\leq 2\%$ (research placeholder); otherwise escalate. **One outcome-blind ECG per patient.** Split fold 9 for calibration, two-lead threshold selection and rescue selection.



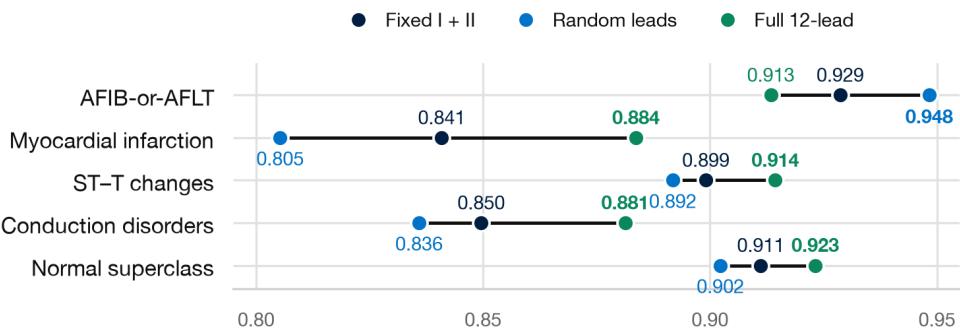
Statistical safeguard: exact one-sided binomial bounds, corrected across 101 thresholds × four policy families [3]. Policies frozen before test export.

04 / RESULTS · EXPLORATORY, SINGLE SEED

2,198 test ECGs · 1,904 patient representatives

Performance depends on the diagnosis.

AUROC by training strategy · higher is better



4 / 5

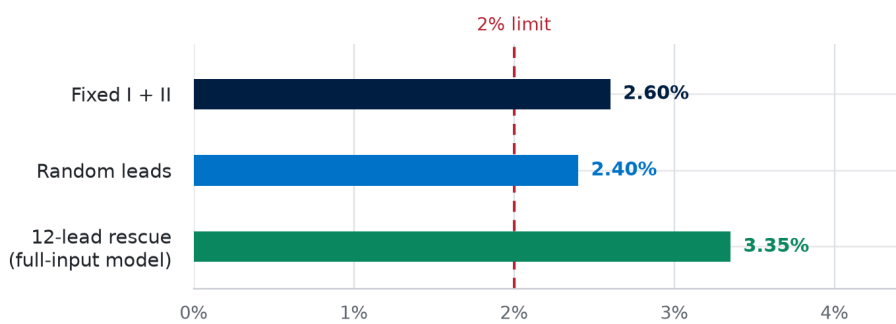
Fixed beats random

0.948

AFIB/AFLT · random leads

Ranking did not enable automation.

Validation upper error bound · lower is better



0%

Automated coverage

100%

12-lead acquisition

100%

Expert referral

05 / RELAXED LIMITS · MORE AUTOMATION, MORE MISSED CASES

Post-hoc sensitivity · same models, patients and 95% confidence

Risk limit	Training	Ruled out at 2 leads	Escalated → Expert	Missed / 118 AFIB-AFLT	Error in ruled-out
3%	Fixed I + II	1,400 (73.5%)	504 (26.5%)	10 / 118 (8.5%)	10 / 1,400 (0.71%)
3%	Random leads	1,604 (84.2%)	300 (15.8%)	13 / 118 (11.0%)	13 / 1,604 (0.81%)
5%	Fixed I + II	1,586 (83.3%)	318 (16.7%)	22 / 118 (18.6%)	22 / 1,586 (1.39%)
5%	Random leads	1,704 (89.5%)	200 (10.5%)	25 / 118 (21.2%)	25 / 1,704 (1.47%)

At 5%, more patients are ruled out, but 22–25 of 118 AFIB/AFLT cases are missed. Twelve-lead rescue accepted no further cases.

N = 1,904 patients. The relaxed limits are post-hoc sensitivity analyses, not clinical targets.

MAIN FINDING

Useful discrimination alone cannot justify automated rule-out.

06 RESEARCH CONTRIBUTION

A reliability map with abstention.

The policy reports automation and misses together under frozen, diagnosis-specific limits [2].

Proof bottleneck: rescue fits saw 47–115 patients; 2% proof needs ≥ 446 error-free accepts. Calibration data, not accuracy, is the bottleneck.

07 LIMITS & NEXT EXPERIMENTS

Feasibility evidence, not clinical validation.

One seed, previously inspected test data and sparse subgroups limit certainty. Hospital ECGs do not reproduce wearable recordings.

Complete seeds 43/44, then test a frozen policy on an untouched external cohort and real two-lead devices.

[1] DATASET

Wagner et al. PTB-XL. *Scientific Data* 7, 154 (2020). doi:10.1038/s41597-020-0495-6

[2] SELECTIVE PREDICTION

Geifman & El-Yaniv. Selective Classification for Deep Neural Networks. *NeurIPS* (2017).

[3] RISK CONTROL

Angelopoulos et al. Learn then Test. arXiv:2110.01052 (2021).

[4] AI USE

AI (ChatGPT) was used to run model performance experiments and generate html for poster baseline.